

Fundamental limits for thermodynamic control with quantum feedback

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Joint work with Gilad Gour (Technion) and Mark M. Wilde (Cornell)



[arXiv:2503.09012](https://arxiv.org/abs/2503.09012)

Overview

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Thermodynamics

- Second law
 - convertibility between states
 - work cost / extractable work
- Resource theory (operational machinery)
 - macroscopic (asymptotic / average)
 - microscopic (single-shot)

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A: **No.**

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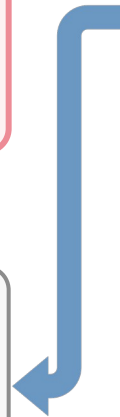
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a classical feedback controller

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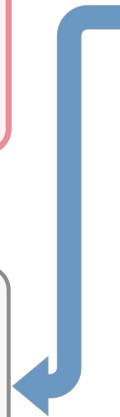
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Coherent quantum feedback

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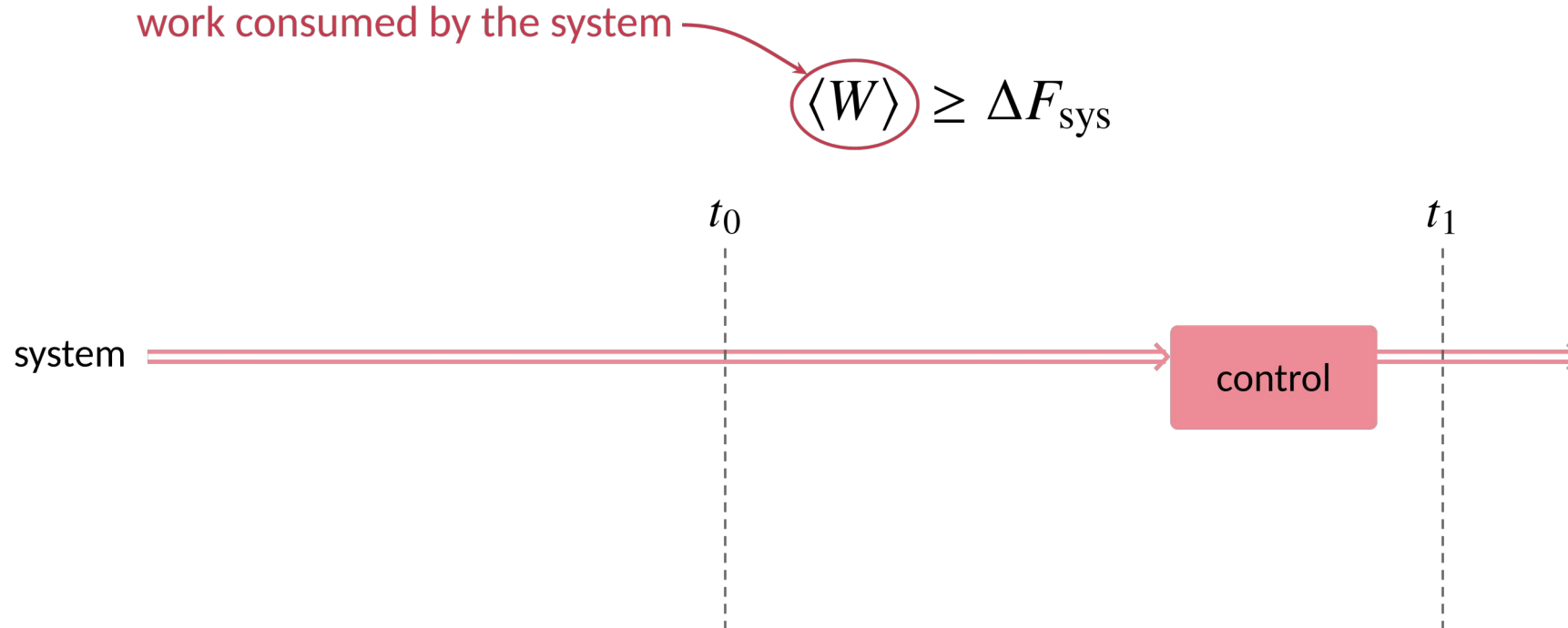
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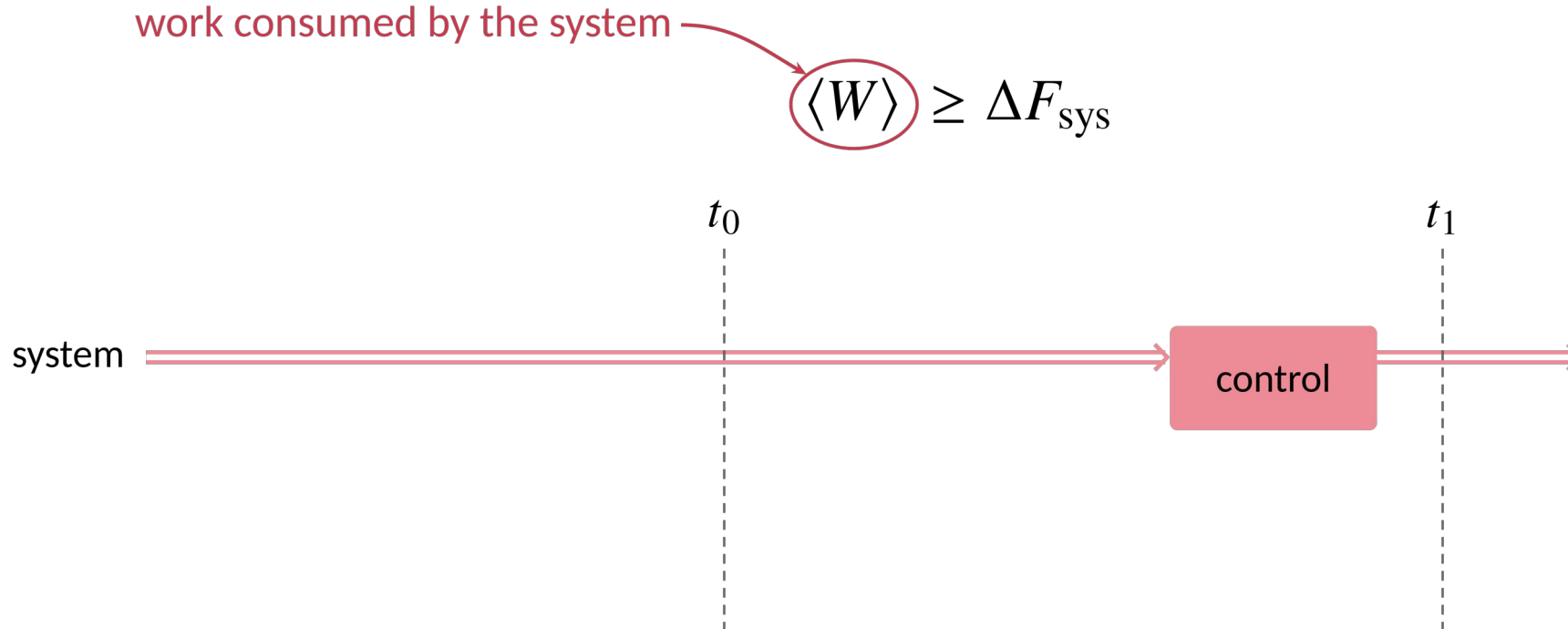
Thermodynamics & feedback control

Second law: the average work consumed by a system is at least its change in (nonequilibrium) free energy



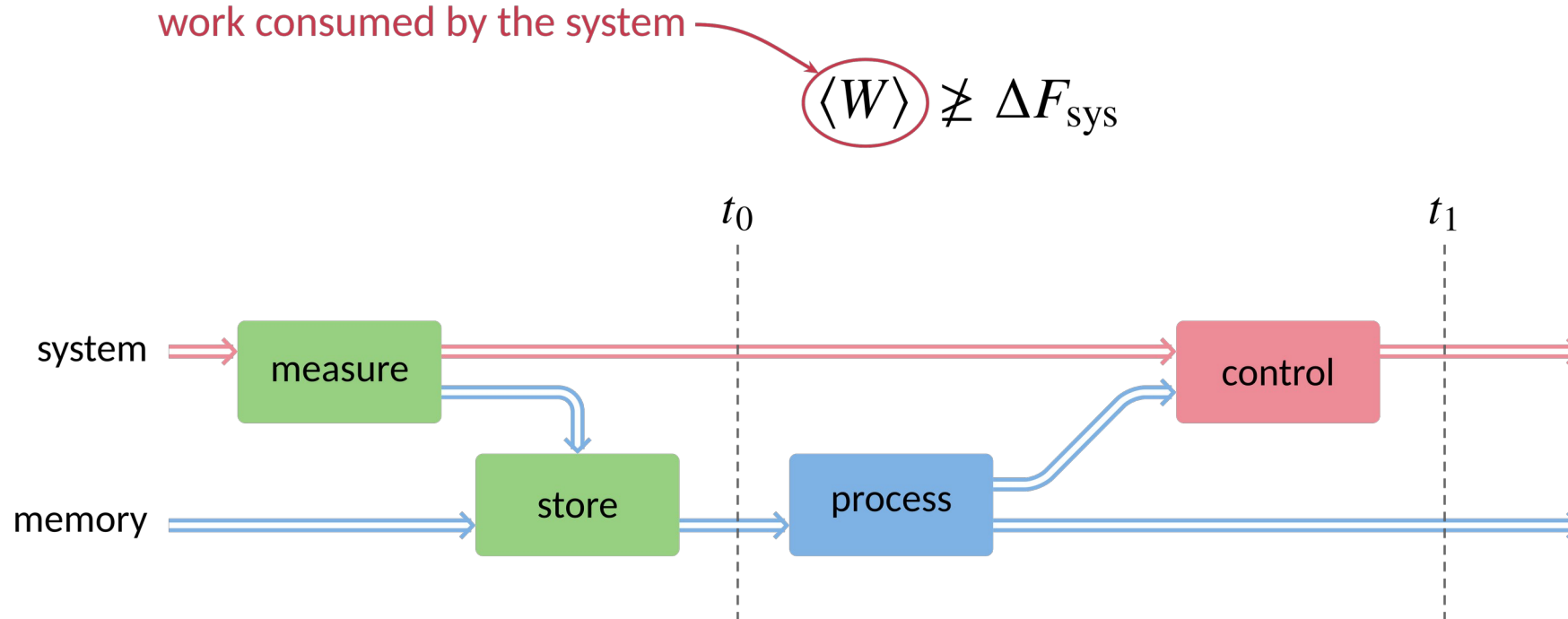
Thermodynamics & feedback control

Maxwell's demon, who holds side information, can violate the second law **locally**



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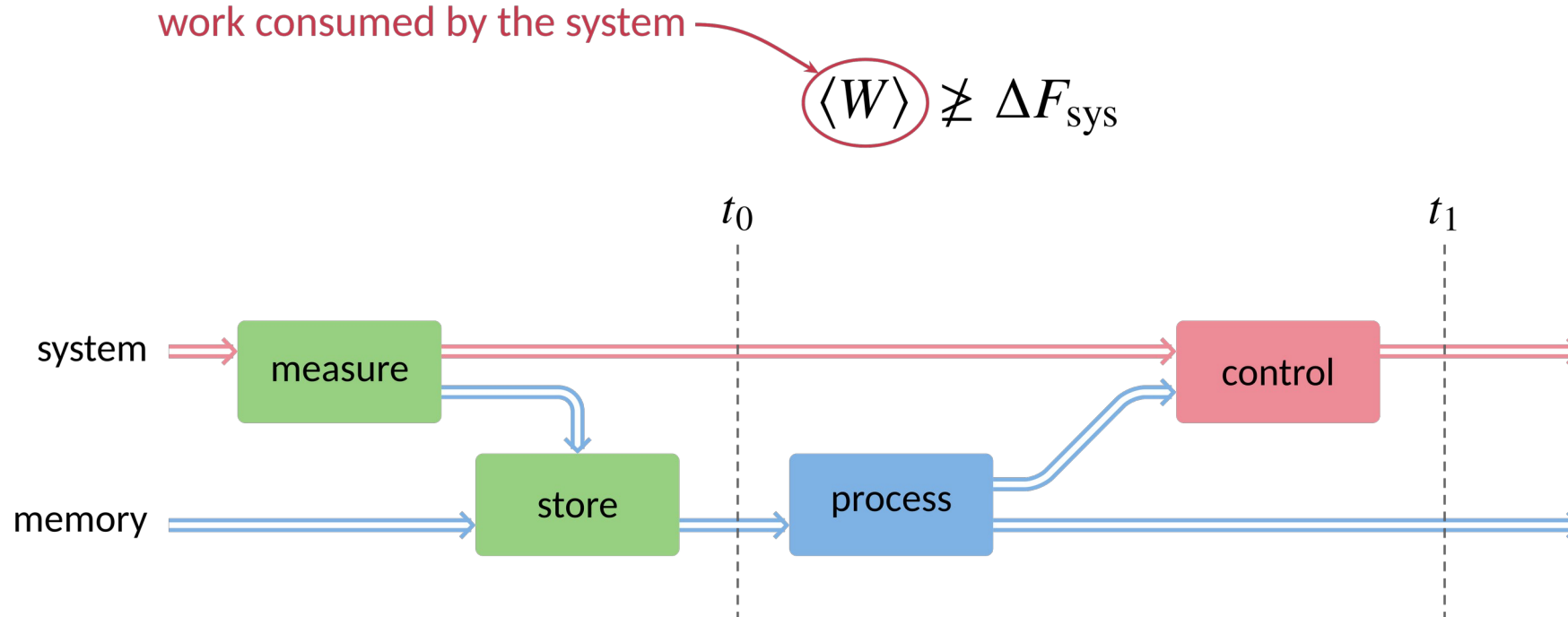
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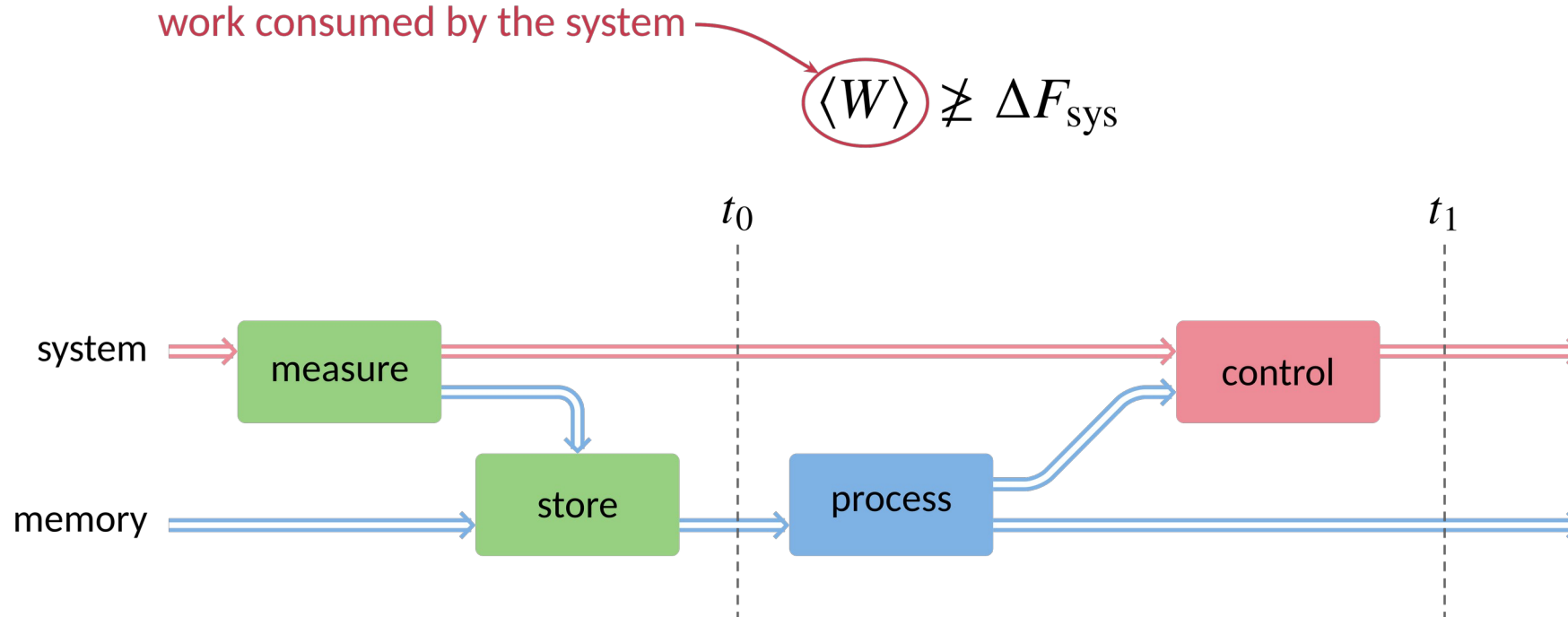
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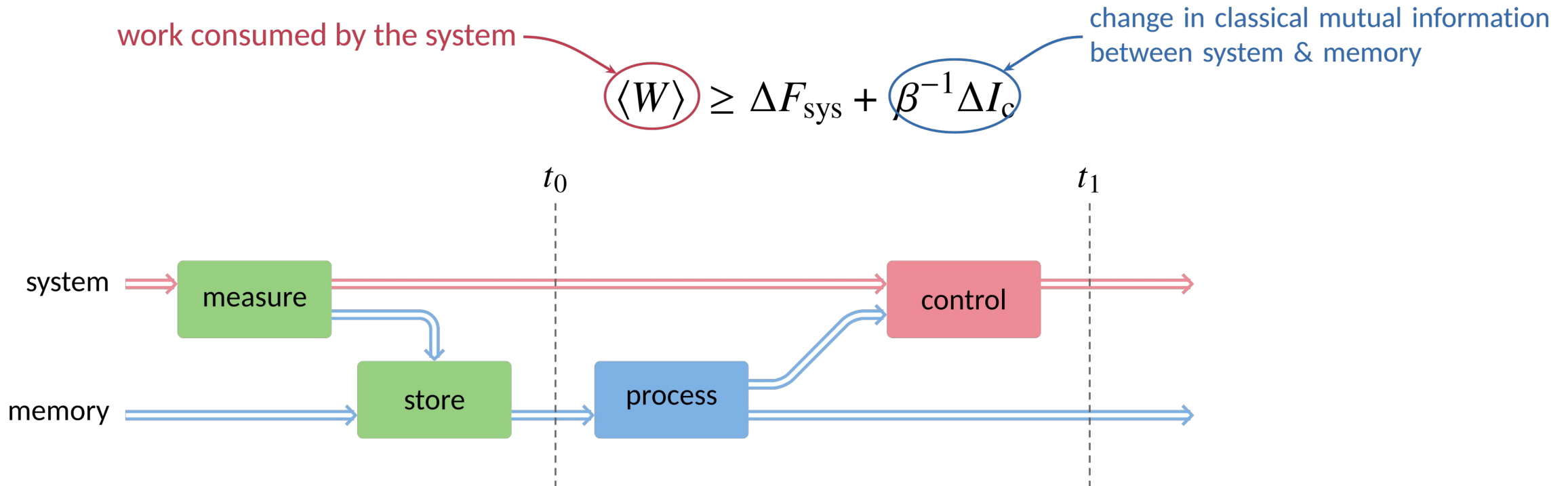
- because of correlations between the system and the demon's memory
- the violation is limited: “second law with (classical) feedback” [Parrondo et al. 2015]



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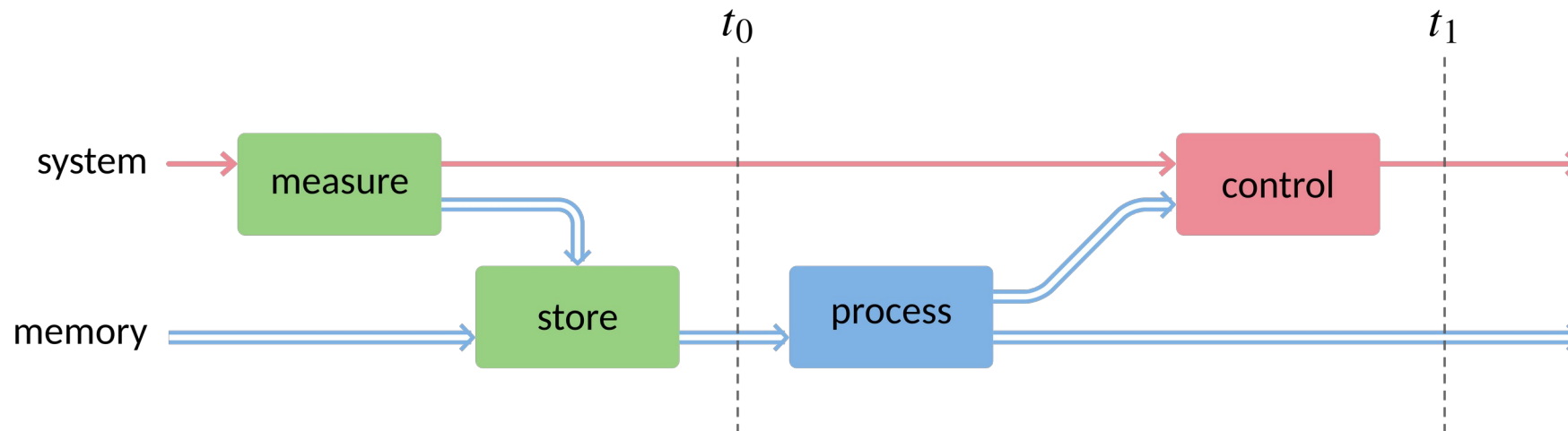
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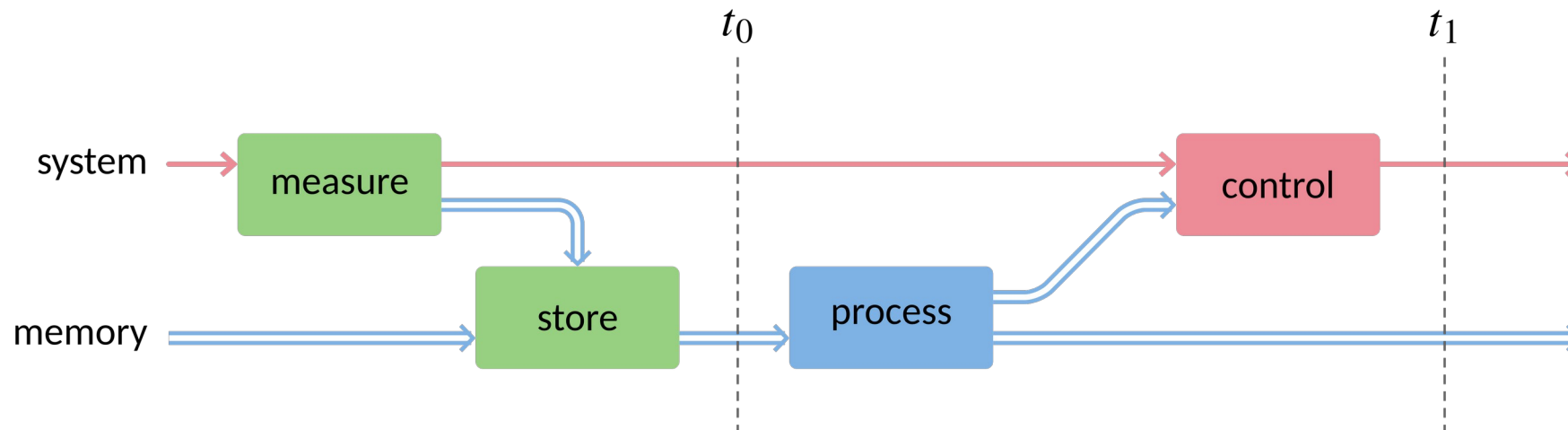
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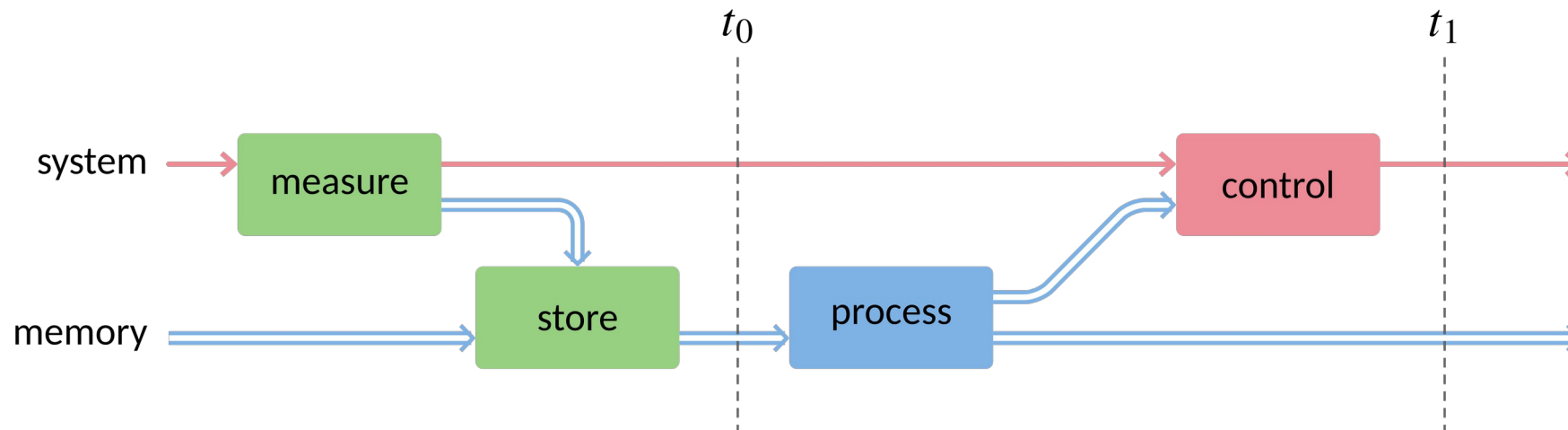
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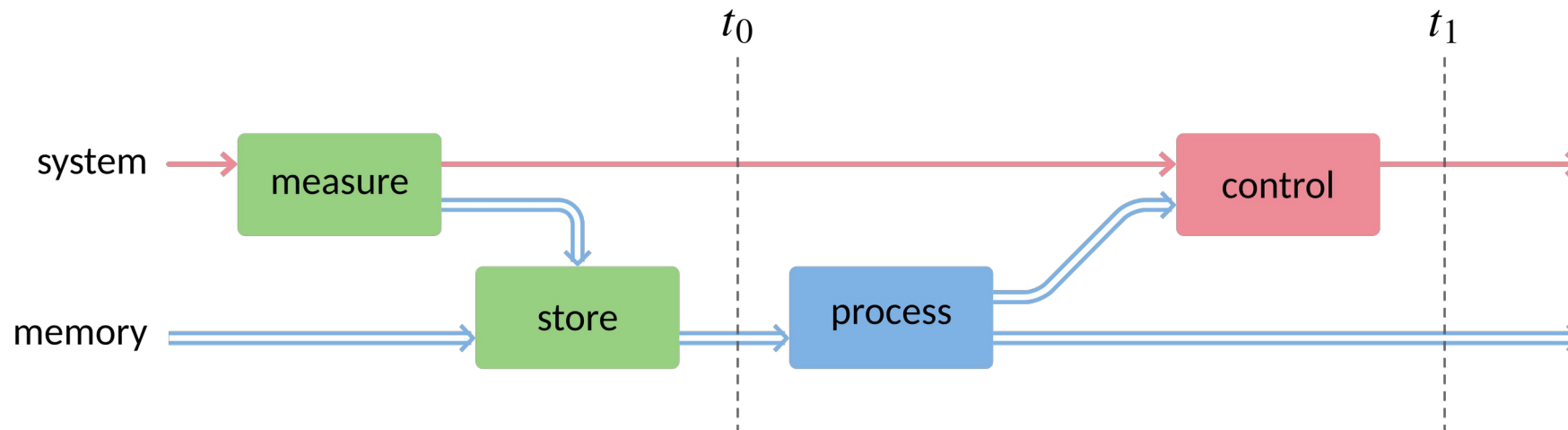
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- side information & feedback are incoherent (a classical demon)
 - the existing “second law with (classical) feedback” is not universal: it could be violated by coherent (quantum) feedback



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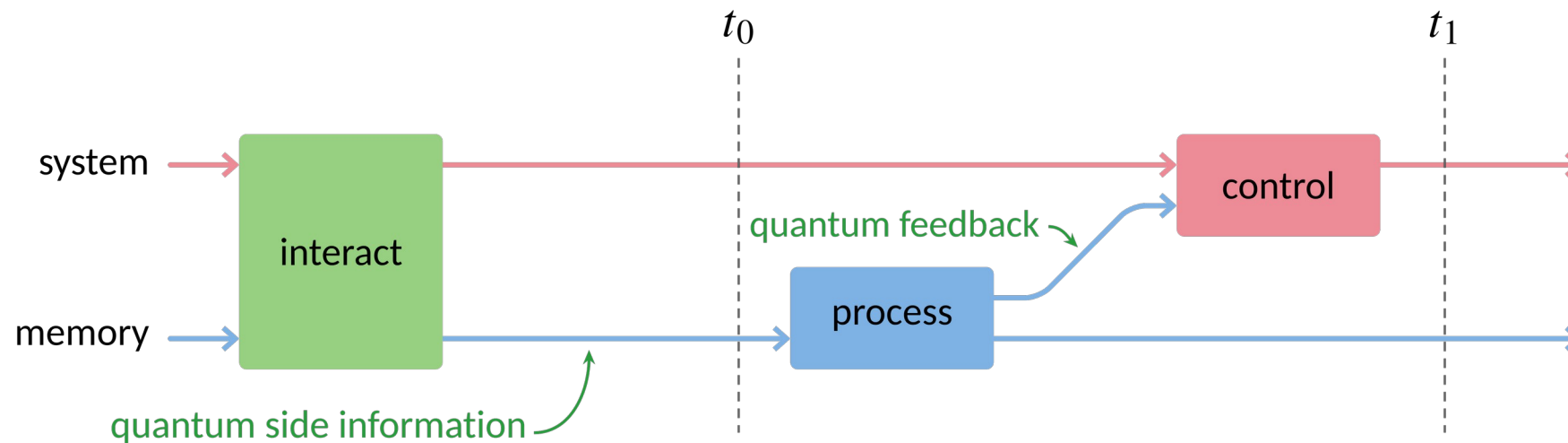
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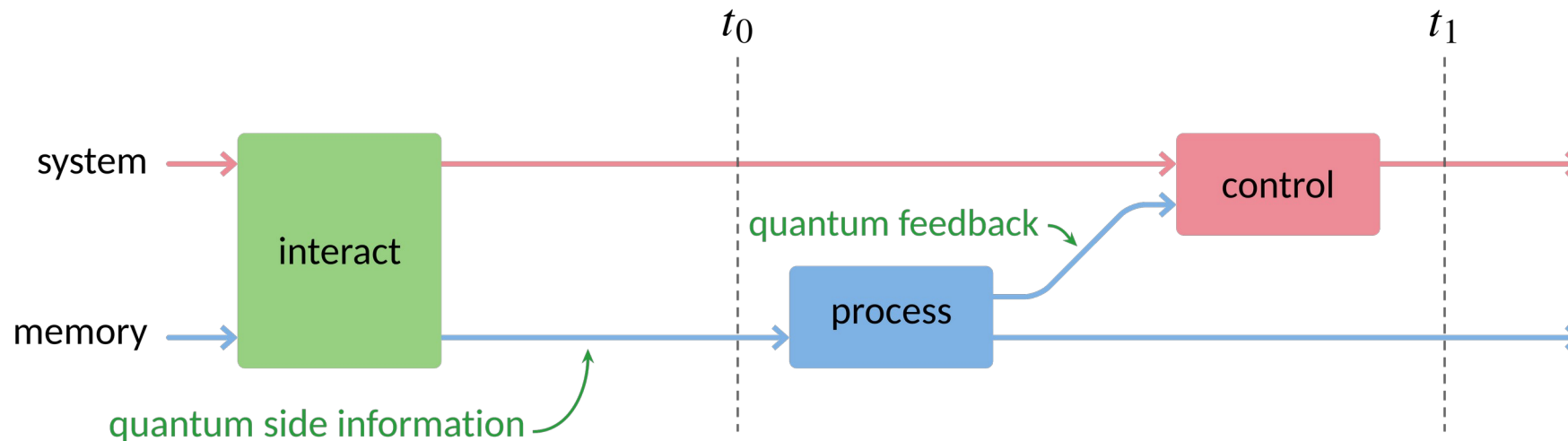
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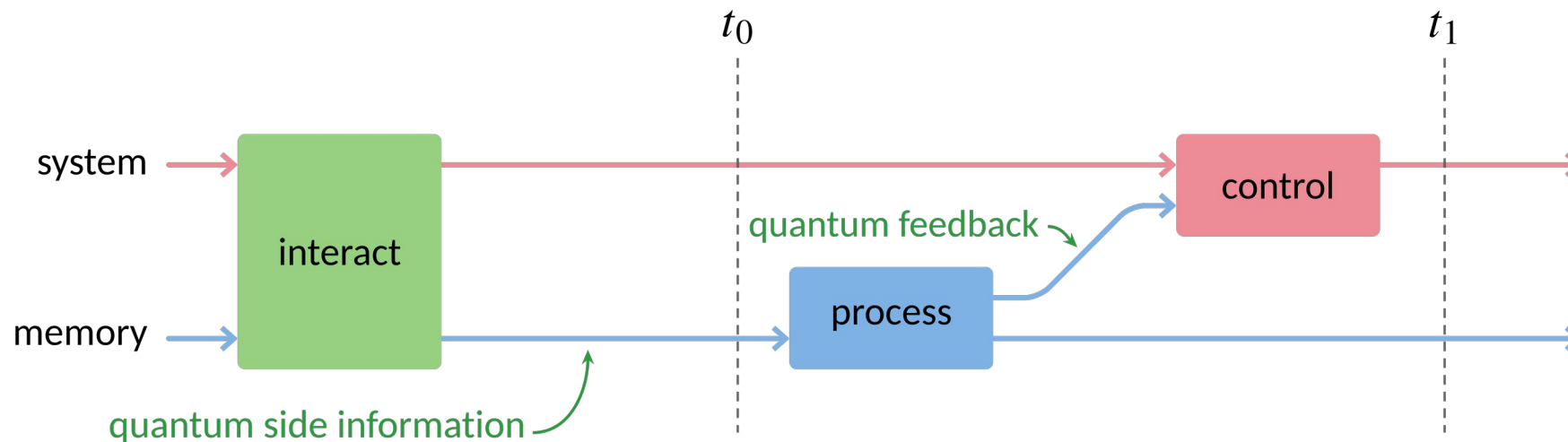
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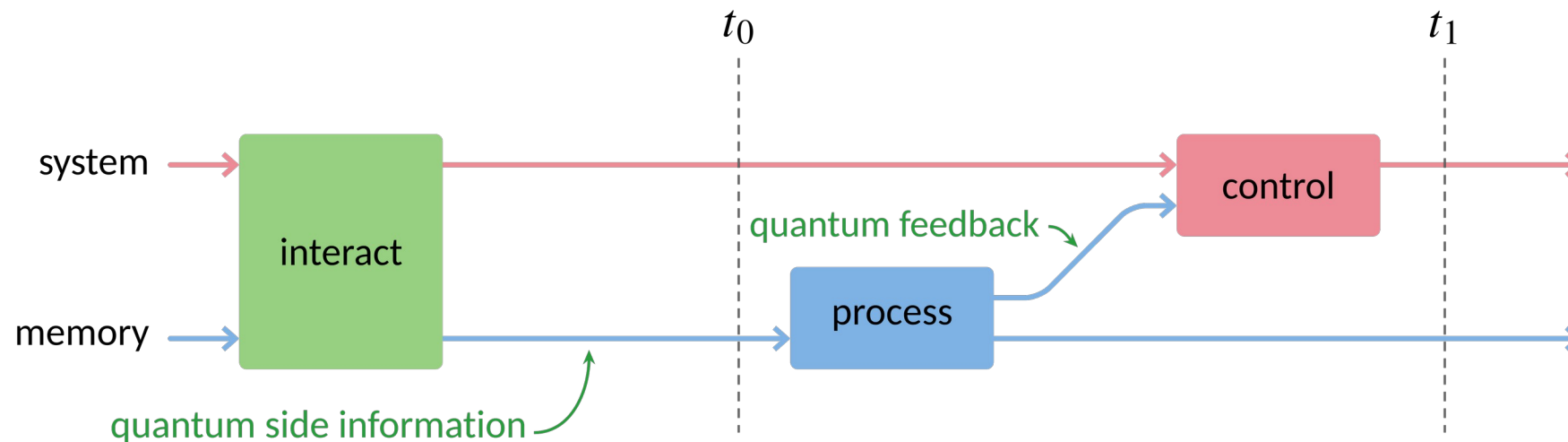
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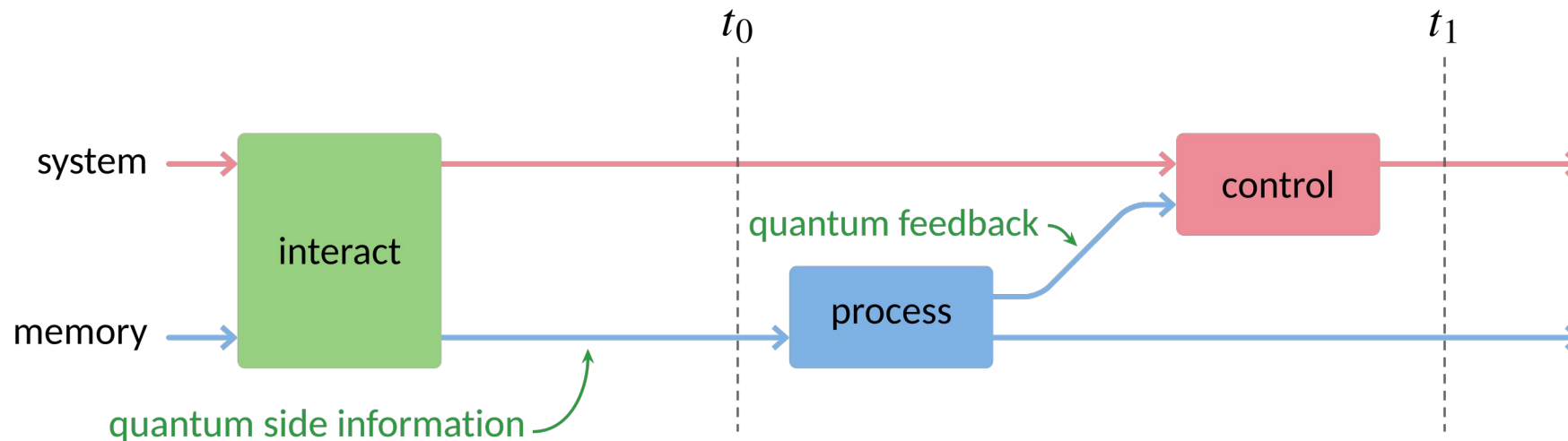
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In the conventional picture of quantum feedback control, sensors perform measurements on the system, a classical controller processes the results of the measurements, and actuators supply semiclassical potentials to alter the behavior of the quantum system. In this picture, the sensors tend to destroy coherence in the process of making measurements, and although the controller can use the actuators to act coherently on the quantum system, it is processing and feeding back classical information. This paper proposes an alternative method for quantum feedback control, in which the sensors, controller, and actuators are quantum systems that interact coherently with the system to be controlled. In this picture, the controller gets, processes, and feeds back *quantum* information. Controllers that operate using such quantum feedback loops can perform tasks such as entanglement transfer that are not possible using classical feedback. Necessary and sufficient conditions are presented for Hamiltonian quantum systems to be controllable and observable using both classical and quantum feedback.

Operational machinery

Resource theory of thermal nonequilibrium in the presence of quantum side information

Free operations

In thermodynamics, the resource is work **expendable on the system**

Free operations: feedback-control schemes that can be performed **without injecting work** into the system

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- a larger set than the truly free operations;
but that is fine for no-go limits: if they hold for these permissive free operations, then they hold for the truly free operations as well

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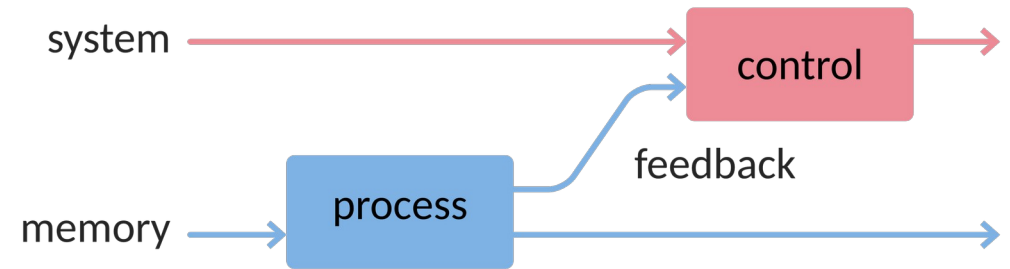
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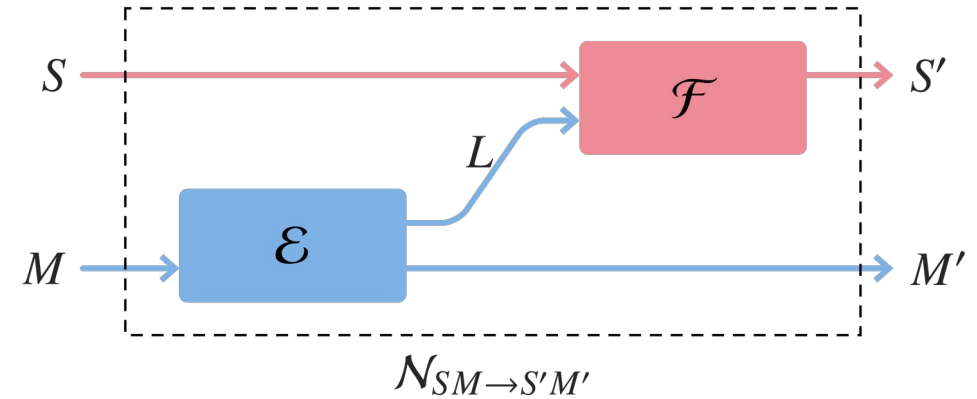
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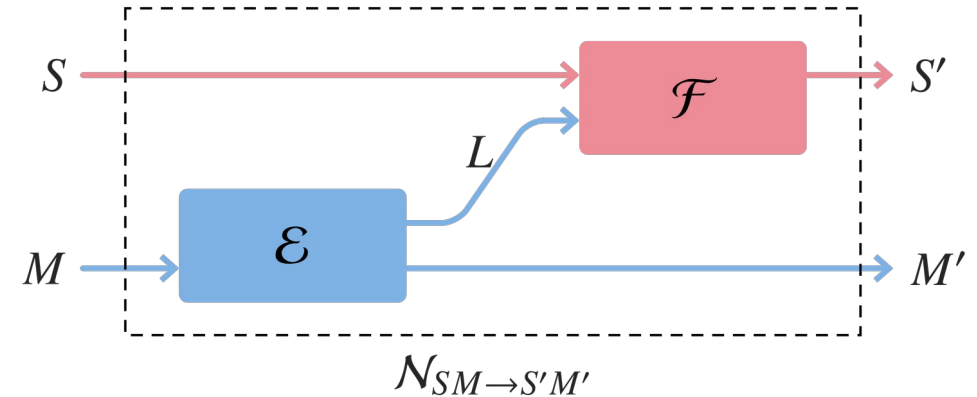
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- ② Gibbs-preserving on the system if no side information is supplied (Kelvin's principle)

$$\mathcal{F}_{SL \rightarrow S'}[\gamma_S \otimes \rho_L] = \gamma'_{S'} \quad \forall \rho_L$$

Gibbs states $\gamma \propto e^{-\beta \hat{H}}$

feedback (L) only supplies information, not work



Free operations: QFGOs

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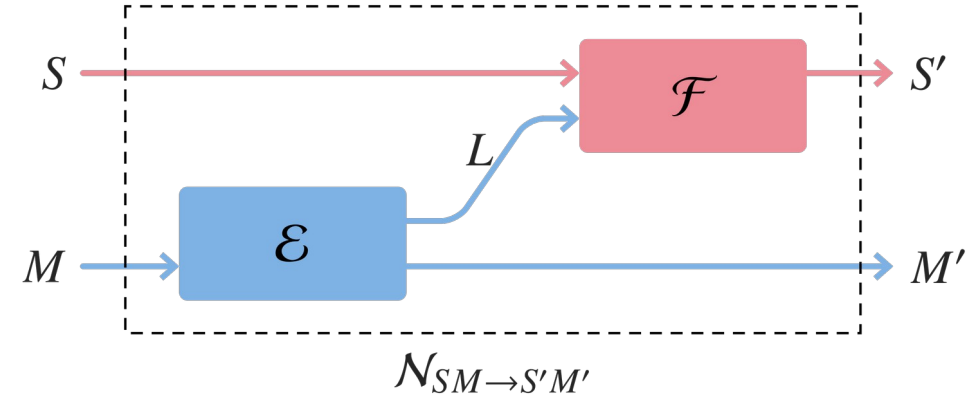
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⇒ Quantum-feedback-assisted Gibbs-preserving operations (**QFGOs**)



Quantifying work

Pure qubits as the currency of work [Szilárd 1929]

- one pure qubit ($|0\rangle\langle 0|$, $\hat{H} = 0$) $\rightleftharpoons \beta^{-1} \ln 2$ of work
- one pure qudit of dimension d $\rightleftharpoons \beta^{-1} \ln d$ of work

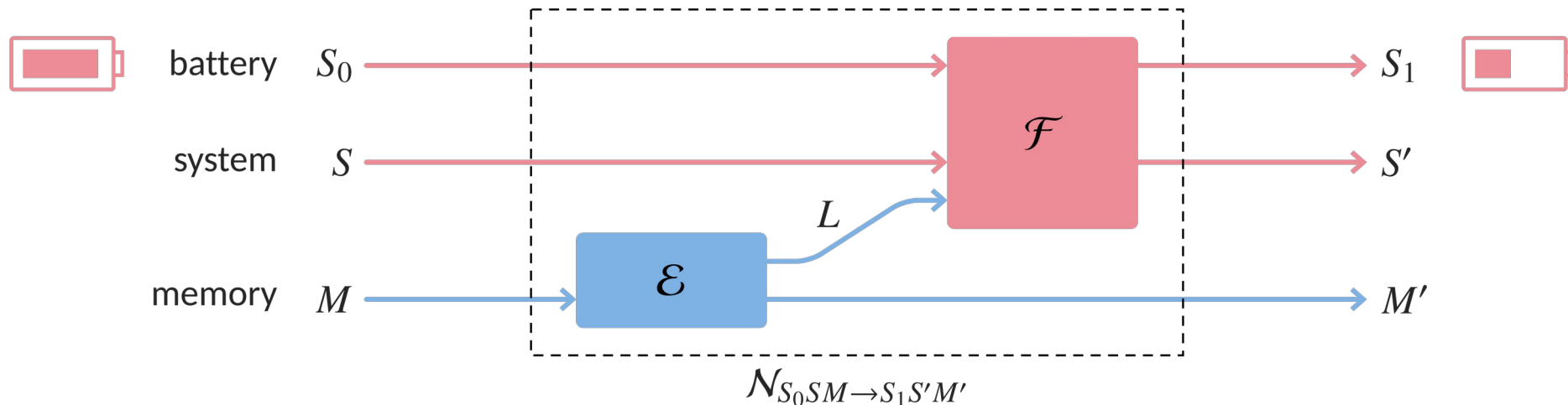
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Battery model [Faist, Dupuis, Oppenheim & Renner 2015]

- put the system alongside a battery S_0 , a pure qudit
- system & battery jointly undergo a QFGO (no external work supply)
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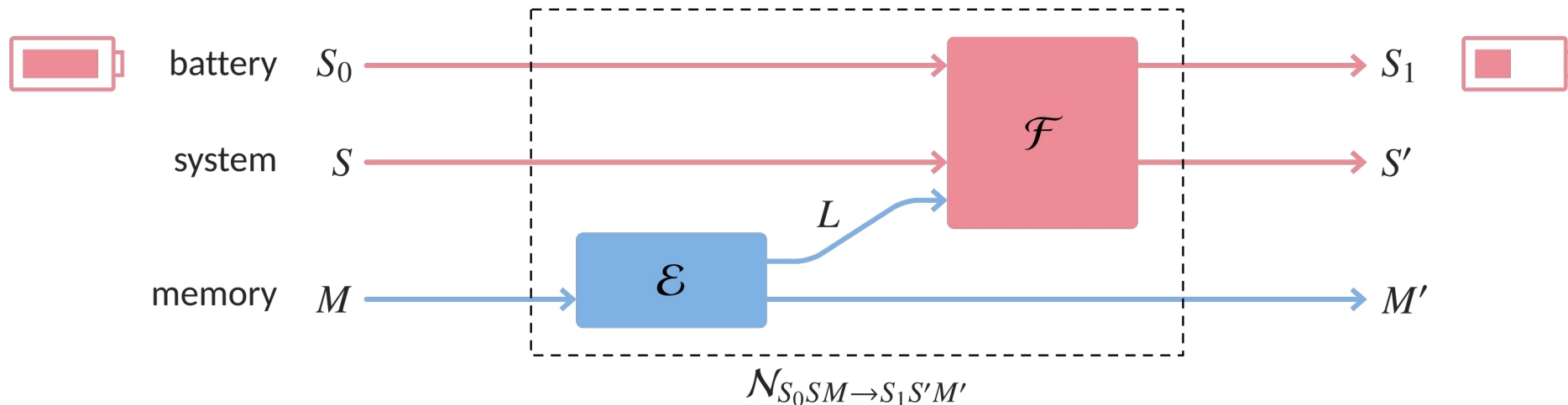
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$\Rightarrow$ work consumed by the system = decrease in battery charge = $\beta^{-1} (\ln |S_0| - \ln |S_1|)$ ($|S|$: dimension of S)



Definition of work cost

Task: transform an initial into a target (state of system & memory, Gibbs state of system)

$$(\rho_{SM}, \gamma_S) \longrightarrow (\rho'_{S'M'}, \gamma'_{S'})$$

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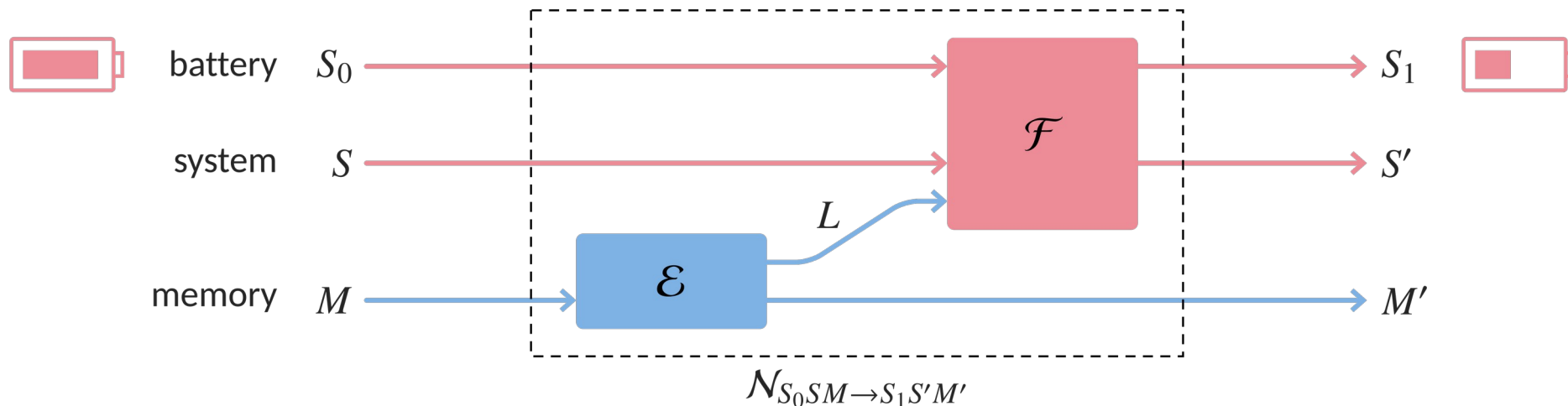
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Work cost, allowing error ε and treating QFGOs as free:

$$W^\varepsilon((\rho_{SM}, \gamma_S) \rightarrow (\rho'_{S'M'}, \gamma'_{S'})) := \inf_{\substack{S_0, S_1, \\ \mathcal{N} \in \text{QFGO}}} \left\{ \beta^{-1} (\ln |S_0| - \ln |S_1|) : \mathcal{N}[|0\rangle\langle 0|_{S_0} \otimes \rho_{SM}] \approx_\varepsilon |0\rangle\langle 0|_{S_1} \otimes \rho'_{S'M'} \right\}$$

↖ in trace distance

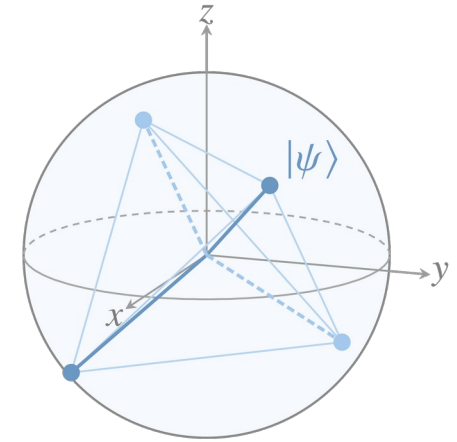


Example: quantum feedback beats classical feedback

A system S correlated with a memory M ; all Hamiltonians trivial

- S : classical, 4 states $s \in \{0, 1, 2, 3\}$, uniformly mixed (thermal)
- M : one qubit holding the record $\hat{\sigma}_s |\psi\rangle$ ($\hat{\sigma}_0 = \mathbb{1}$; $\hat{\sigma}_{1,2,3}$: Pauli)
with $|\psi\rangle\langle\psi| = \frac{1}{2}(\hat{\sigma}_0 + \frac{1}{\sqrt{3}}(\hat{\sigma}_1 + \hat{\sigma}_2 + \hat{\sigma}_3))$

S alone is thermal: no work is extractable without feedback



Bloch vectors of the 4 records:
vertices of a regular tetrahedron

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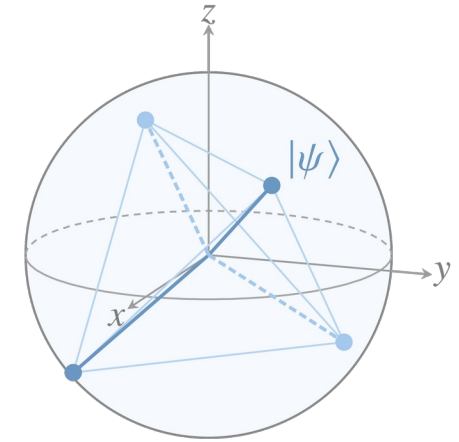
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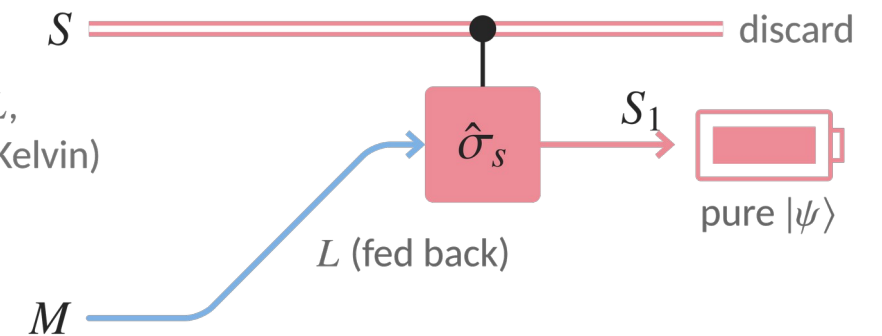
Work extractable from S with feedback from M :

quantum feedback  $\beta^{-1} \ln 2$ (single shot)

scheme at right; a QFGO: with S thermal & independent of L , the controlled Pauli twirls L into $\mathbb{1}/2$, so it creates no work (Kelvin)



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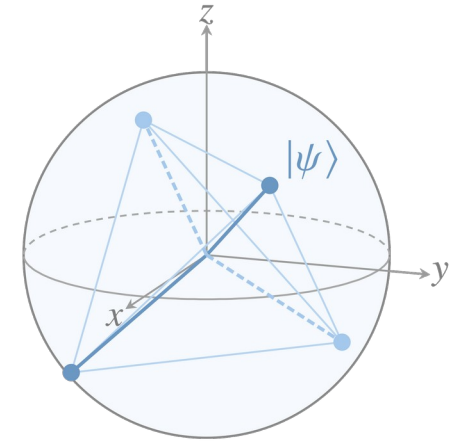
scheme at right; a QFGO: with S thermal & independent of L , the controlled Pauli twirls L into $\mathbb{1}/2$, so it creates no work (Kelvin)

classical feedback  $\leq \beta^{-1} \ln \frac{4}{3}$ (even asymptotically)

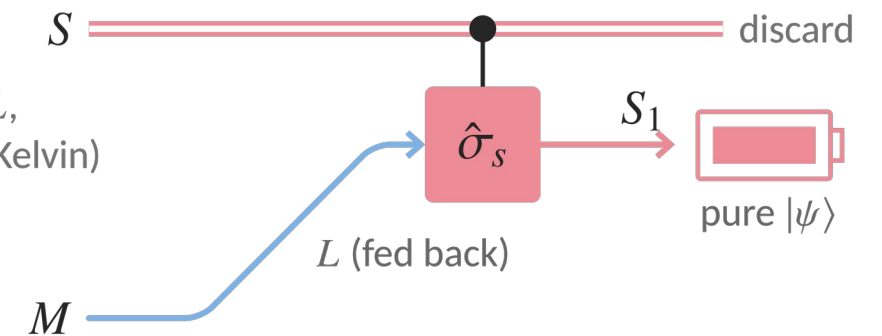
measure $M \rightarrow X$; second law with classical feedback:

$$W_{\text{extr}} \leq \beta^{-1} I(S; X) \leq \beta^{-1} I_{\text{acc}} = \beta^{-1} \ln \frac{4}{3}$$

(I_{acc} : accessible information of the 4 records) [Holevo 1973]



Bloch vectors of the 4 records:
vertices of a regular tetrahedron



Main results

Single-shot & asymptotic work costs with quantum feedback

Task 1: state formation with quantum feedback

Starting from thermal equilibrium, create a target state (ρ_{SM}, γ_S) at minimum work cost

- Application: writing quantum data from an external memory into a device (e.g., a hard drive)
 - the device can end up entangled with the memory: impossible with classical feedback

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Definition: single-shot work of formation under QFGOs

$$W_{\text{form}}^{\varepsilon}(\rho_{SM}, \gamma_S) := W^{\varepsilon}(\emptyset \rightarrow (\rho_{SM}, \gamma_S))$$

no system & memory to start with (equivalently, S thermal)

Task 1: state formation with quantum feedback

Theorem 1. Single-shot & asymptotic work of formation under QFGOs:

$$W_{\text{form}}^{\varepsilon}(\rho_{SM}, \gamma_S) = \beta^{-1} I_{\max}^{\uparrow, \varepsilon}(\rho_{SM} \| \gamma_S) \quad \forall \varepsilon \in [0, 1]$$
$$\lim_{n \rightarrow \infty} \frac{1}{n} W_{\text{form}}^{\varepsilon}(\rho_{SM}^{\otimes n}, \gamma_S^{\otimes n}) = \beta^{-1} I(\rho_{SM} \| \gamma_S) \quad \forall \varepsilon \in (0, 1)$$

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Smoothed max-generalized mutual information (GMI): **SDP computable**

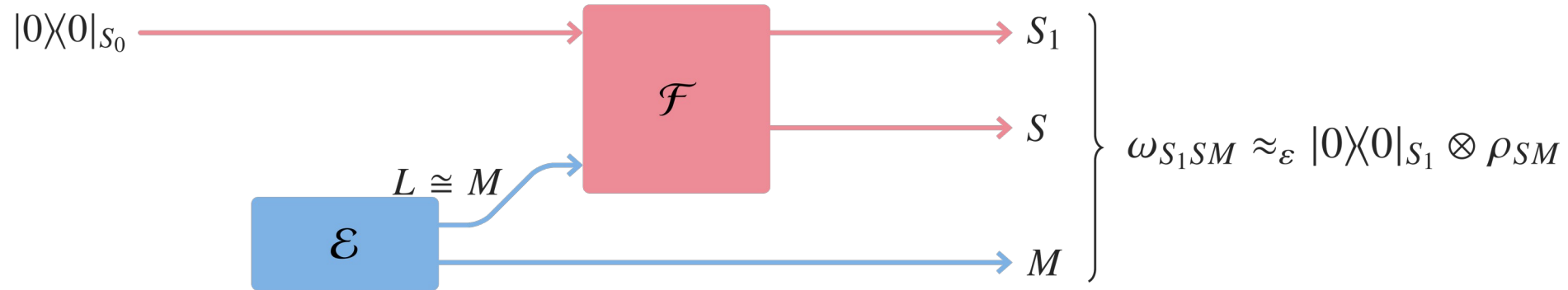
$$I_{\max}^{\uparrow, \varepsilon}(\rho_{SM} \| \gamma_S) := \inf_{t, \tau_{SM}, \sigma_M} \left\{ \ln t : \tau_{SM} \leq t \gamma_S \otimes \sigma_M, \tau_M \leq \sigma_M, \tau_{SM} \approx_{\varepsilon} \rho_{SM} \right\}$$

 τ_{SM} : subnormalized state; σ_M : state

Umegaki GMI:

$$I(\rho_{SM} \| \gamma_S) := D(\rho_{SM} \| \gamma_S \otimes \rho_M) = \underbrace{D(\rho_S \| \gamma_S)}_{\text{nonequilibrium of } S} + \underbrace{I(S; M)_{\rho}}_{\text{correlation (quantum mutual information)}}$$

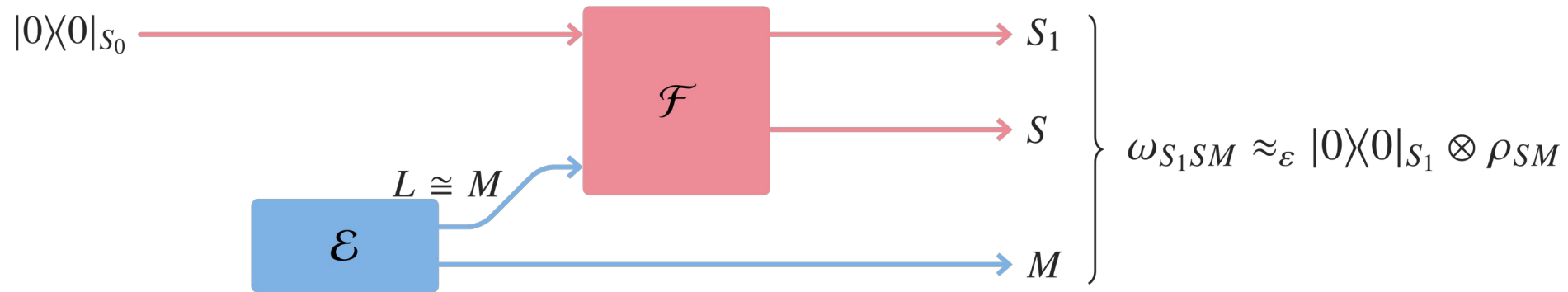
Task 1: optimal protocol



Protocol

- $\mathcal{E}$: prepare $|M| \omega_M^{1/2} \Phi_{LM} \omega_M^{1/2}$ (Φ : maximally entangled) and feed $L \cong M$ back
- $\mathcal{F}$: measure the battery S_0 ; outcome 0 $\Rightarrow$ apply $\mathcal{F}_{L \rightarrow S_1 S}^{(0)}$ (output $\omega_{S_1 S M}$); otherwise $\Rightarrow$ apply $\mathcal{F}_{L \rightarrow S_1 S}^\perp$

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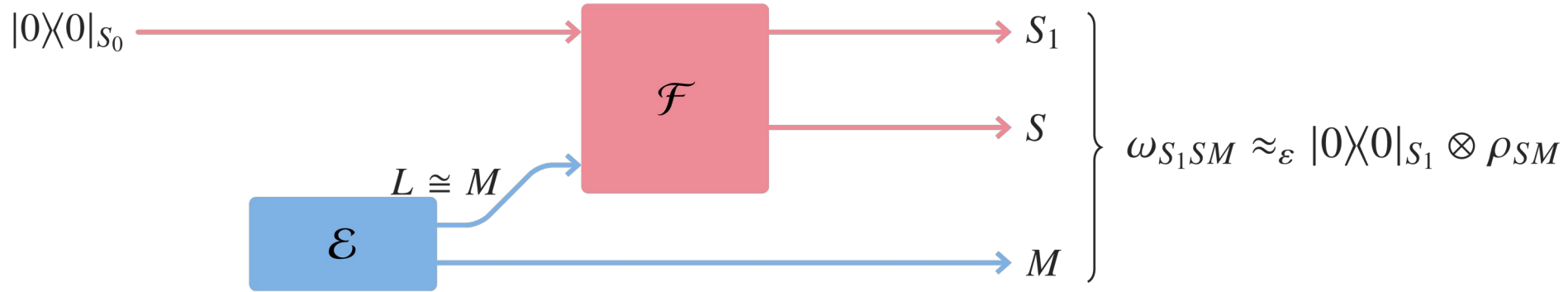
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Why it works

- Kelvin: a thermal battery gives outcome 0 only w.p. $1/|S_0|$; $\mathcal{F}^\perp$ makes the average output $\frac{\mathbb{1}_{S_1}}{|S_1|} \otimes \gamma_S$ for any L , so the scheme is a QFGO

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- $\mathcal{F}^\perp$ is CPTP as long as $|S_0| \geq e^{D_{\max}(\omega_{S_1 S M} \| \frac{\mathbb{1}_{S_1}}{|S_1|} \otimes \gamma_S \otimes \omega_M)}$. So, for a suitable S_1 and $\omega_{S_1 S M}$, the minimum cost is

$$\beta^{-1}(\ln |S_0| - \ln |S_1|) \approx \beta^{-1}(D_{\max}(\omega_{S_1 S M} \| \frac{\mathbb{1}_{S_1}}{|S_1|} \otimes \gamma_S \otimes \omega_M) - \ln |S_1|) = \beta^{-1} I_{\max}^{\uparrow, \epsilon}(\rho_{SM} \| \gamma_S)$$

Task 2: work extraction with quantum feedback

Starting from (ρ_{SM}, γ_S) , extract as much work as possible

- Prior work: classical feedback, asymptotic [Morris, Lami & Adesso 2019]

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Definition: single-shot extractable work under QFGOs

$$W_{\text{extr}}^{\varepsilon}(\rho_{SM}, \gamma_S) := -W^{\varepsilon}((\rho_{SM}, \gamma_S) \rightarrow \emptyset)$$

 the initial state is used up

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Theorem 2. Single-shot & asymptotic extractable work under QFGOs:

$$W_{\text{extr}}^{\varepsilon}(\rho_{SM}, \gamma_S) = \beta^{-1} I_{\min}^{\downarrow, \varepsilon}(\rho_{SM} \| \gamma_S) \quad \forall \varepsilon \in [0, 1]$$
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Smoothed min-GMI: **SDP computable**

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a test: hypothesis testing between ρ_{SM} and $\{\gamma_S \otimes \sigma_M : \sigma_M \text{ a state}\}$

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Erasing S ($\hat{H}_S = 0$), i.e., resetting it to $|0\rangle\langle 0|_S$, with quantum feedback: a single-shot Landauer principle

$$W_{\text{erase}}^\varepsilon = \beta^{-1} \ln |S| - W_{\text{extr}}^\varepsilon(\rho_{SM}, \frac{\mathbb{1}_S}{|S|}) = \beta^{-1} H_{\max}^{\uparrow, \varepsilon}(S|M)_\rho$$

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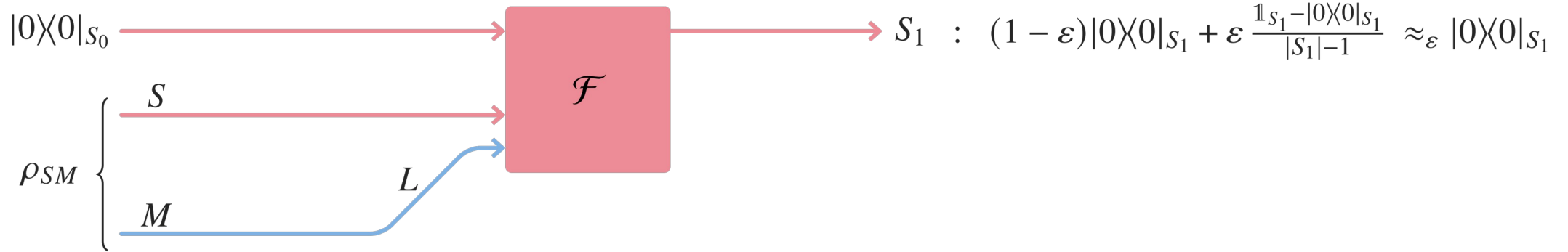
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can be negative: cooling rather than heating

cf. [del Rio et al. 2011]

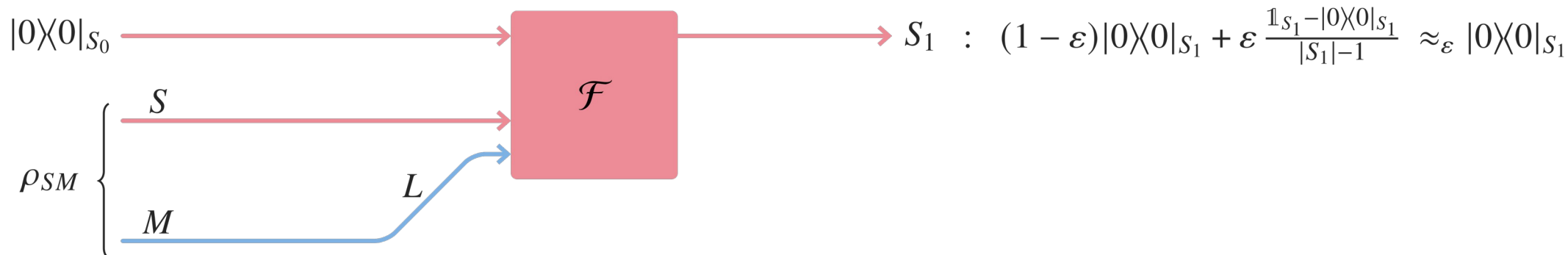
Task 2: optimal protocol



Protocol

- $\mathcal{E}$: feed the memory back unchanged ($M \rightarrow L$)
- $\mathcal{F}$: test S_0SL with Λ_{SL} (accepts ρ_{SM} w.p. $1 - \varepsilon$); accept $\Rightarrow |0\rangle\langle 0|_{S_1}$, reject $\Rightarrow \frac{\mathbb{1}_{S_1} - |0\rangle\langle 0|_{S_1}}{|S_1| - 1}$

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Protocol

- $\mathcal{E}$: feed the memory back unchanged ($M \rightarrow L$)
- $\mathcal{F}$: test S_0SL with Λ_{SL} (accepts ρ_{SM} w.p. $1 - \epsilon$); accept $\Rightarrow |0\rangle\langle 0|_{S_1}$, reject $\Rightarrow \frac{\mathbb{1}_{S_1} - |0\rangle\langle 0|_{S_1}}{|S_1| - 1}$

Why it works

- Kelvin: if S is thermal, it accepts w.p. $\leq \|\text{Tr}_S[\Lambda_{SL}\gamma_S]\|_{\infty}$ for any L ; a QFGO if $\frac{1}{|S_1|} \approx \frac{1}{|S_0|} \|\text{Tr}_S[\Lambda_{SL}\gamma_S]\|_{\infty}$
- work extracted: $\beta^{-1} (\ln |S_1| - \ln |S_0|) \approx -\beta^{-1} \ln \|\text{Tr}_S[\Lambda_{SM}\gamma_S]\|_{\infty} = \beta^{-1} I_{\min}^{\downarrow, \epsilon}(\rho_{SM} \parallel \gamma_S)$

Extracting work = telling ρ_{SM} apart from “thermal S , any feedback”

Generalized second law with quantum feedback

Theorem 3. Asymptotic work cost under QFGOs, $\forall \varepsilon \in (0, 1)$:

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Conditional nonequilibrium free energy: [Bera et al. 2017]

$$F(S|M)_\rho := \text{Tr}[\hat{H}_S \rho_S] - \beta^{-1} H(S|M)_\rho = F(S)_\rho + \beta^{-1} I(S; M)_\rho$$

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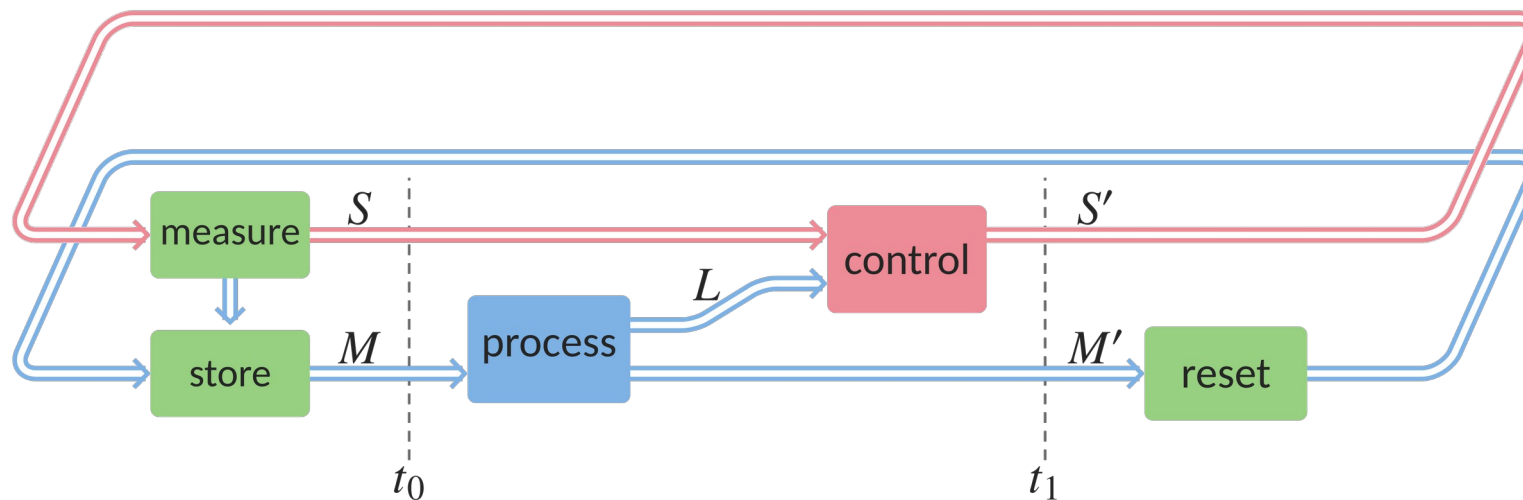
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↑
change in **quantum** mutual information (classical before)

Consistency with the traditional second law

Maxwell's demon violates the second law only **locally**

- no global violation over a closed cycle: the work gained through feedback is paid back elsewhere (measurement, reset) [Bennett 1982] [Sagawa & Ueda 2009] [Minagawa et al. 2025]

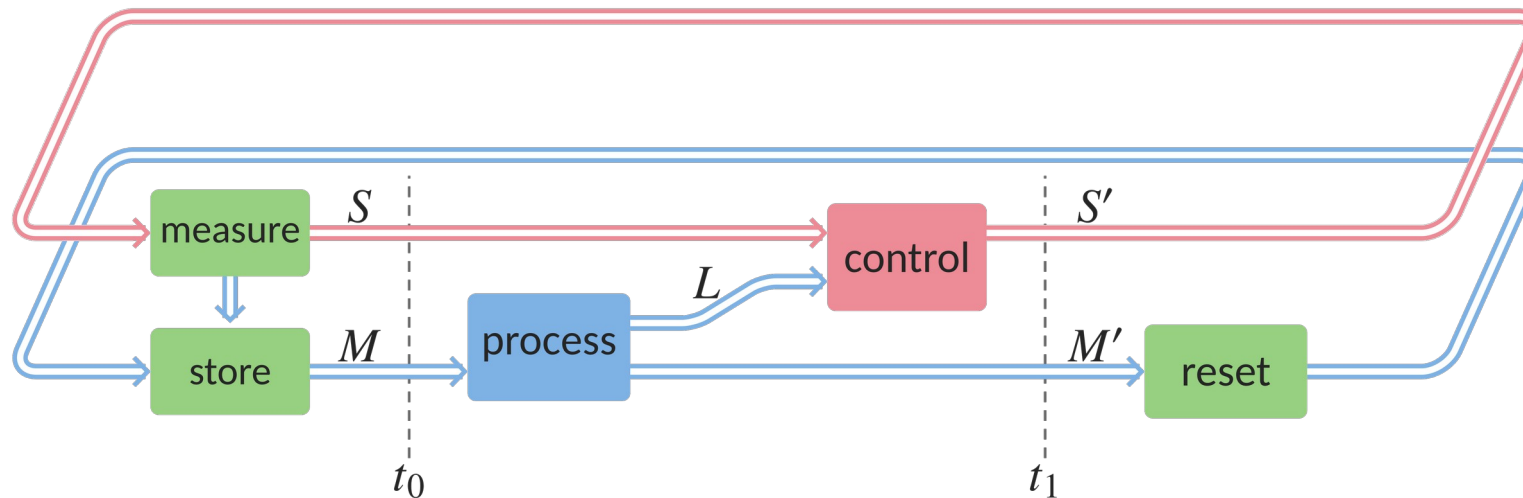


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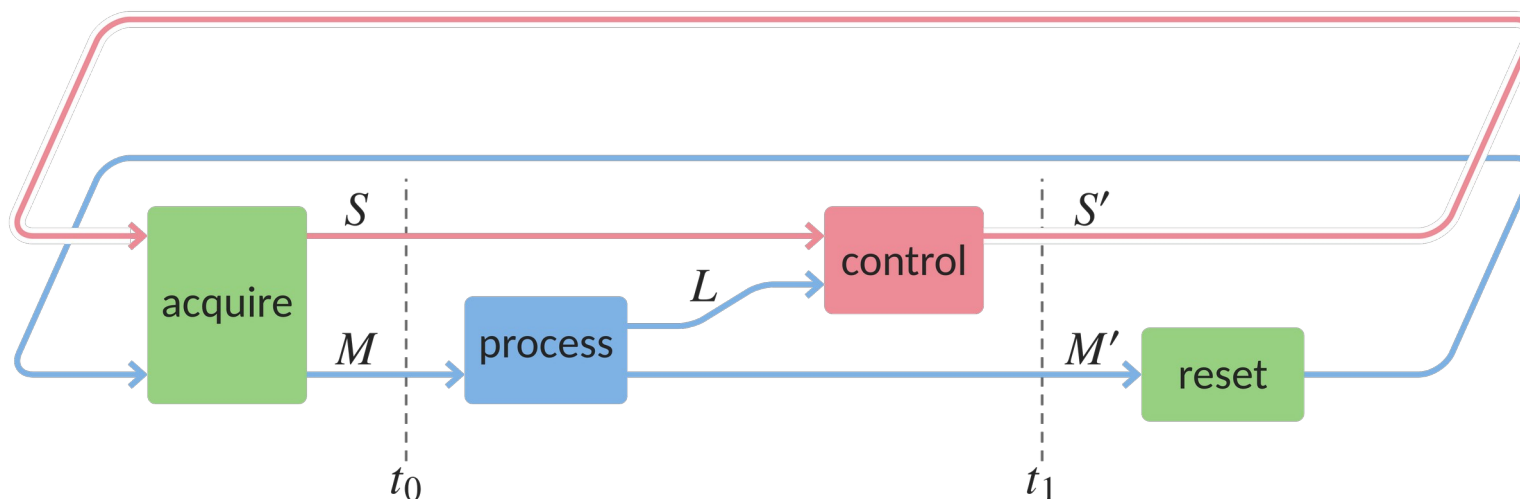
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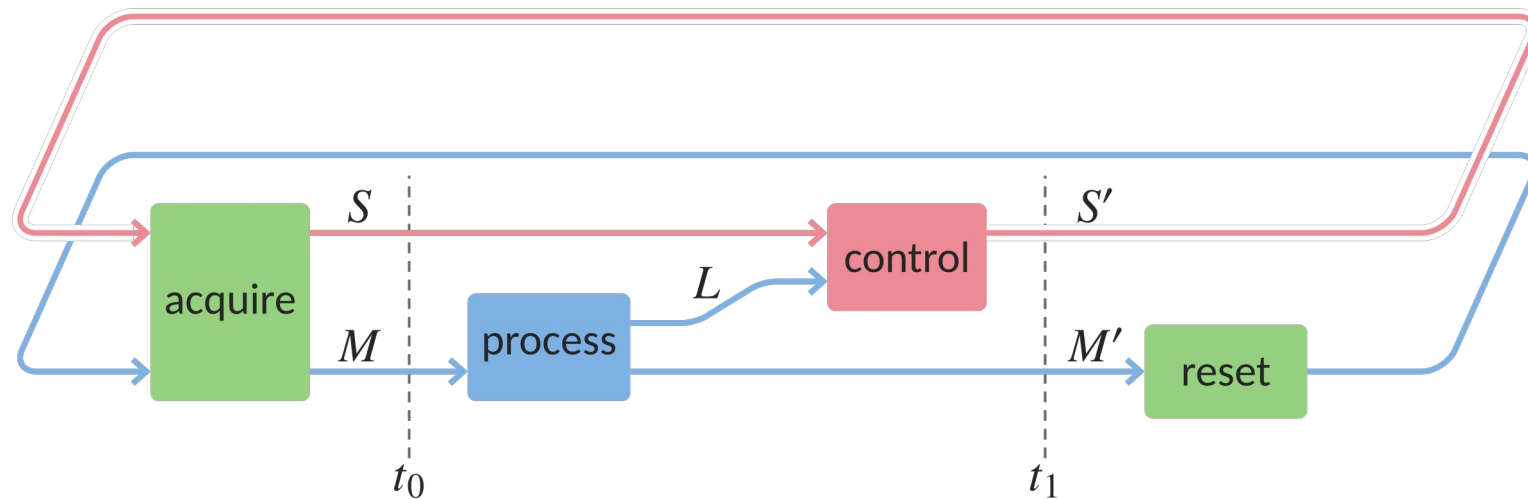
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- most general cycle: quantum memory throughout; acquisition by any interaction (not a measurement); quantum processing & feedback



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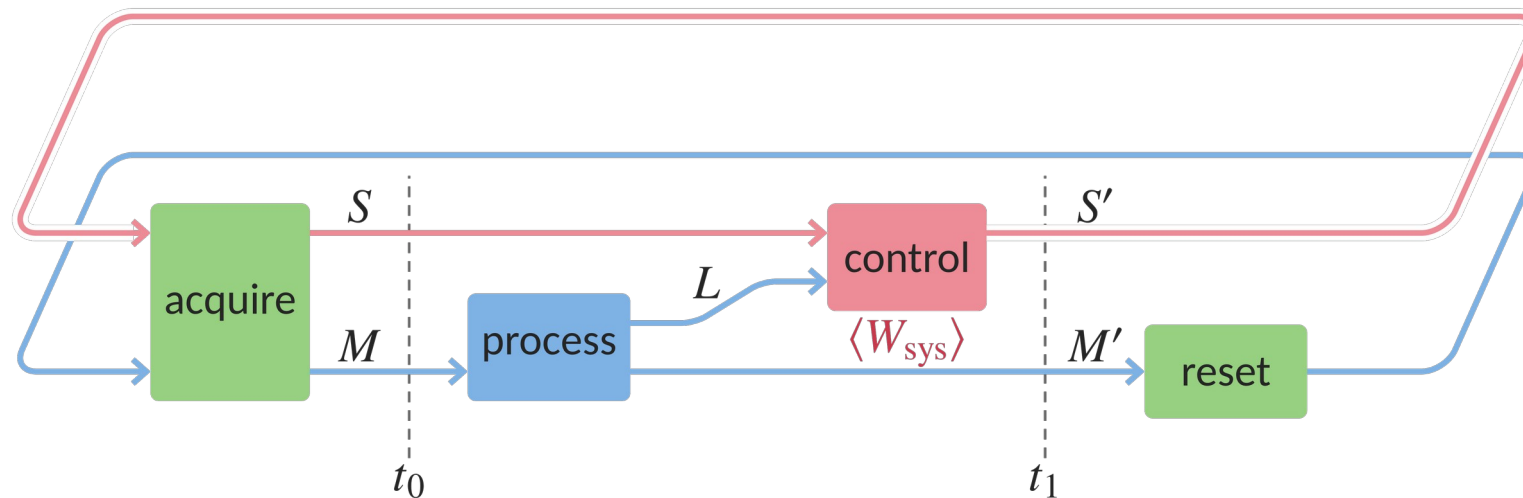


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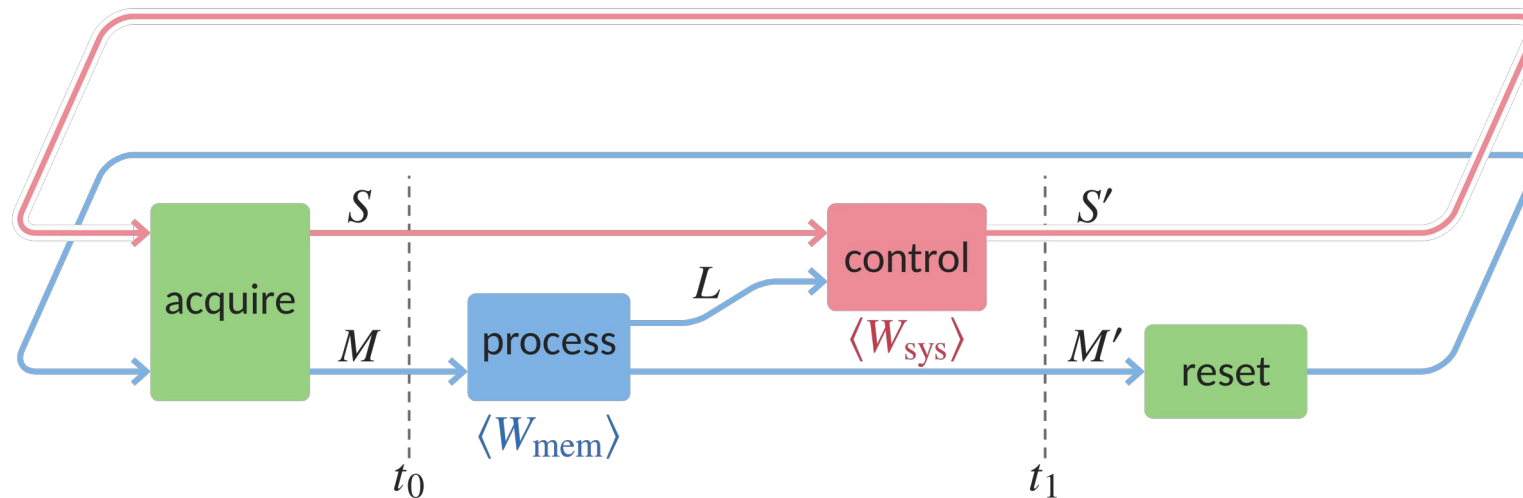
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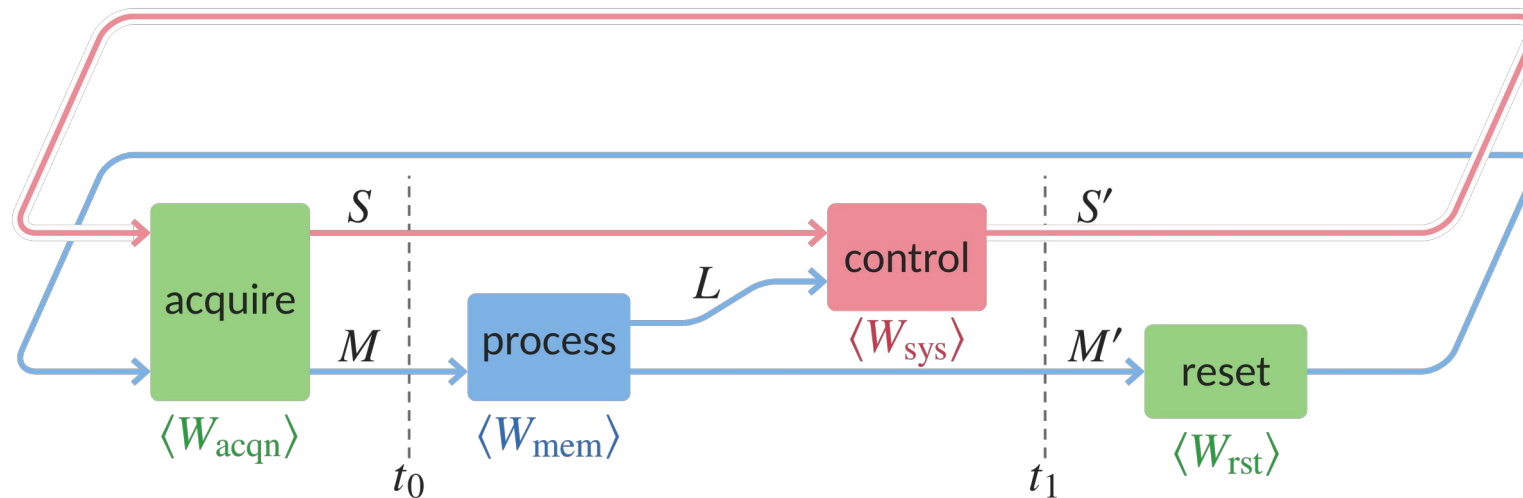
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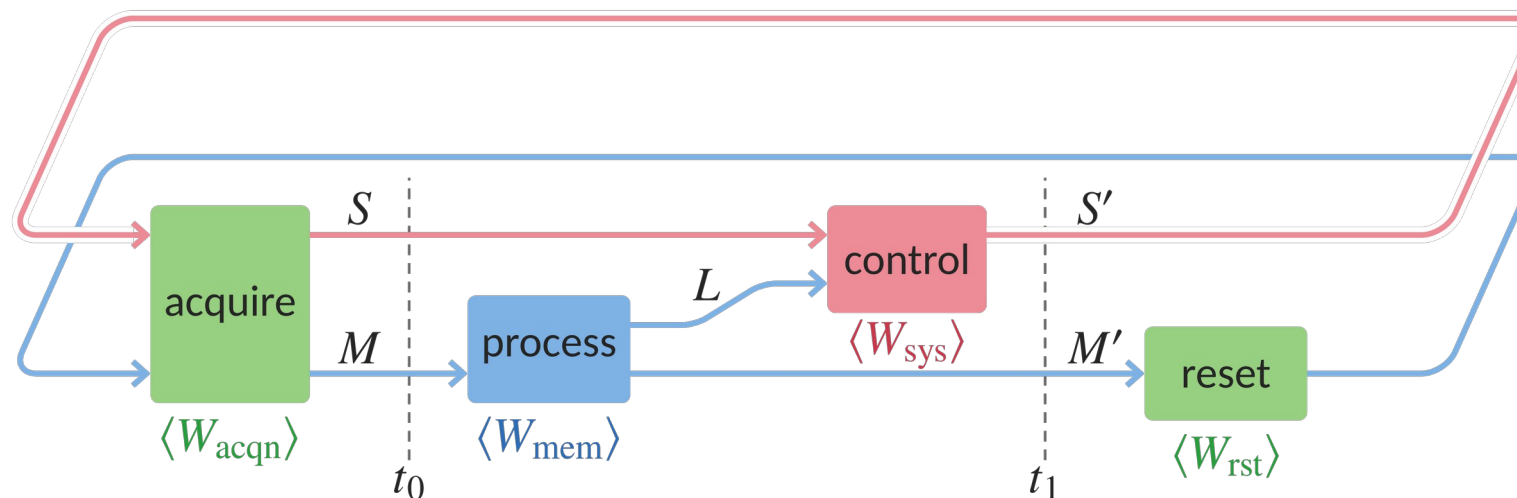
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$$\Rightarrow \langle W_{\text{sys}} \rangle + \langle W_{\text{mem}} \rangle + \langle W_{\text{acqn}} \rangle + \langle W_{\text{rst}} \rangle \geq 0$$

as the traditional second law predicts



Conclusion

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- Quantum feedback can strictly beat classical feedback (example: $\beta^{-1} \ln 2$ vs. $\leq \beta^{-1} \ln \frac{4}{3}$, even for a classical system, without entanglement)

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- Quantum error correction (QEC) as feedback control
 - a round of QEC is a feedback-control cycle
 - measurement-based / measurement-free QEC = classical / coherent feedback [Heußen, Locher & Müller 2024]
 - ...

